
Multimodal sample
image
62 F
smoker, 30 py
cough, wt loss
RUL ..... seen
nodule
text
Partial views & subsets
α:color
β:filter
γ:silhouette
mask a
mask b
mask c
each view = $x_i^{(v)}$
View-specific adapter
view $x_i^{(v)}$
CLIP
ViT-B/32
frozen
Adapter
g_v
trainable
32-dim adapter output
$h^{(v)}$
s_1 {α, β}
s_2 {β, γ}
s_3 {a, b}
s_4 {b, c}
View Subset S_b
Contrastive factor blocks
$h^{(v)}$
content selector
m_b
trainable
content
$c_i^{(b,v)}$
pull
push
$c_i^{(b,v')}$
$c_j^{(b,v')}$
InfoNCE Loss
Representative pairs
$P_i^{(b)} = \text{mean}\, c_i^{(b,v)}$
P_b
differ in b,
similar else
other blocks q
select top-K pairs
MLLM naming & FCI
MLLM
$\Psi \rightarrow \text{name}_j$
predicted factor
GT factor
Nodule Status
lesion
Tobacco History
smoking
Patient Age
age
Patient Sex
gender
smoking
gender
age
diagnosis
Image Brightness